

Non-Atomic L_p Spaces Fail the SPAP

Automated Math Discovery Smoke Test

June 8, 2026

Source and Open Question

This packet concerns Question 2 from:

Luis A. Garcia, Jose Lucas P. Luiz, Vinicius C. C. Miranda, *Norm attainment for multilinear operators and polynomials on Banach Spaces and Banach lattices*, arXiv:2605.12117.

The source location is Section 4, “Open Questions”, Question 2. The question appears on page 28 of the source paper:

Corollary 4.2. *Let $n \in \mathbb{N}$, E be a reflexive Banach lattice and F be a Dedekind complete Banach lattice. Consider the following implications:*

- (1) *Every $A \in \mathcal{L}^+(E^n; F)$ is norm-attaining.*
- (2) *Every $A \in \mathcal{L}^+(E^n; F)$ is weakly sequentially continuous.*
- (3) *Every $P \in \mathcal{P}^+(E^n; F)$ is weakly sequentially continuous.*
- (4) *Every $P \in \mathcal{P}^+(E^n; F)$ is norm-attaining.*

If E or F has the SPAP, then (1) $\iff$ (2) $\Rightarrow$ (3) $\iff$ (4).

In Section 3, we introduced the SPAP and presented examples of Banach lattices that do and do not have this property. So far, however, the only known class of Banach lattices with the SPAP consists of Banach lattices with order continuous norm whose order is given by a basis. This leads to the following question:

Question 2: Are there other examples of Banach lattices with the SPAP besides Banach lattices with order continuous norm whose order is given by a basis?

We showed in Section 3 that all positive bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0, 1])$ are norm-attaining while there are non-norm attaining bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0, 1])$. From this and our results, we obtain that all positive bilinear operators $\ell_4 \times \ell_4 \rightarrow L_1([0, 1])$ are weakly sequentially continuous while there are bilinear operators from $\ell_4 \times \ell_4$ into $L_1([0, 1])$ that are not weakly sequentially continuous. It would be interesting, if we get examples with the domain being non-atomic Banach lattices. In view of this, we leave the following question:

Question 3: Are all positive n -linear forms on $L_{p_1}[0, 1] \times \cdots \times L_{p_n}[0, 1]$ with $\sum_{i=1}^n \frac{1}{p_i} < 1$ weakly sequentially continuous?

The paper defines the sequentially positive approximation property (SPAP) as follows: a Banach lattice E has SPAP if there is a sequence $(S_n)_n \subset B_{\mathcal{F}_+(E;E)}$ such that

$$\lim_{n \rightarrow \infty} \|S_n - \text{id}_E\|(x) = 0$$

for every $x \in E^+$. The authors note that Banach lattices with order continuous norm whose order is given by a basis have SPAP, while $L_2([0, 1])$ and $C([0, 1])$ do not.

Claim

Theorem 1. *Let (Ω, Σ, μ) be a non-atomic finite measure space with $0 < \mu(\Omega) < \infty$, and let $1 \leq p < \infty$. Then $L_p(\mu)$ does not have the SPAP.*

Remark 1. *This is a partial negative result for Question 2. It does not construct a new Banach lattice with SPAP. It rules out the natural non-atomic L_p family, extending the paper's displayed obstruction for $L_2([0, 1])$ to all finite p and all non-atomic finite measure spaces.*

Proof

We first record the weakly-null sign sequence used in the obstruction.

Lemma 1. *There is a sequence $(r_k)_{k \geq 1}$ in $L_\infty(\mu)$ such that $|r_k| = \mathbf{1}_\Omega$ almost everywhere for every k , and $r_k \xrightarrow{w} 0$ in $L_p(\mu)$ for every $1 \leq p < \infty$.*

Proof. Since μ is non-atomic and finite, recursively split each set in a finite partition into two measurable pieces of equal measure. This gives dyadic partitions

$$\Omega = \bigsqcup_{\sigma \in \{0,1\}^m} A_\sigma^{(m)}, \quad \mu(A_\sigma^{(m)}) = 2^{-m} \mu(\Omega),$$

where each $A_\sigma^{(m)}$ is split into $A_{\sigma 0}^{(m+1)} \sqcup A_{\sigma 1}^{(m+1)}$. Let $\mathcal{G}_m$ be the finite sigma-algebra generated by the m th partition, and define

$$r_m = \sum_{\sigma \in \{0,1\}^{m-1}} (\mathbf{1}_{A_{\sigma 0}^{(m)}} - \mathbf{1}_{A_{\sigma 1}^{(m)}}).$$

Then $|r_m| = \mathbf{1}_\Omega$ and $\mathbb{E}(r_m \mid \mathcal{G}_{m-1}) = 0$.

Fix $1 \leq p < \infty$ and let q be the conjugate exponent, with $q = \infty$ when $p = 1$. For $g \in L_q(\mu)$, finite measure implies $g \in L_1(\mu)$. Put $M_m = \mathbb{E}(g \mid \mathcal{G}_m)$. By the martingale convergence theorem, $(M_m)_m$ converges in L_1 to $\mathbb{E}(g \mid \bigvee_m \mathcal{G}_m)$, hence $\|M_m - M_{m-1}\|_1 \rightarrow 0$. Since r_m is $\mathcal{G}_m$ -measurable,

$$\int_\Omega g r_m d\mu = \int_\Omega M_m r_m d\mu = \int_\Omega (M_m - M_{m-1}) r_m d\mu.$$

The last equality uses $\mathbb{E}(r_m \mid \mathcal{G}_{m-1}) = 0$. Therefore

$$\left| \int_\Omega g r_m d\mu \right| \leq \|M_m - M_{m-1}\|_1 \rightarrow 0.$$

This proves $r_m \xrightarrow{w} 0$ in $L_p(\mu)$. □

Assume, for contradiction, that $L_p(\mu)$ has SPAP. Then there is a sequence $(S_n)_n \subset B_{\mathcal{F}_+(L_p; L_p)}$ such that

$$\|S_n - I(\mathbf{1}_\Omega)\| \rightarrow 0.$$

Choose n_0 so that

$$\|S_{n_0} - I(\mathbf{1}_\Omega)\| < \frac{1}{4} \|\mathbf{1}_\Omega\|_p.$$

The operator S_{n_0} has finite rank, hence it is compact. Since $r_k \xrightarrow{w} 0$ in $L_p(\mu)$, compactness gives

$$\|S_{n_0} r_k\|_p \rightarrow 0.$$

For all sufficiently large k ,

$$\|S_{n_0}r_k - r_k\|_p \geq \|r_k\|_p - \|S_{n_0}r_k\|_p > \frac{3}{4} \|\mathbf{1}_\Omega\|_p.$$

On the other hand, for the regular operator $S_{n_0} - I$,

$$|S_{n_0} - I|(|r_k|) \geq |(S_{n_0} - I)r_k|.$$

Because $|r_k| = \mathbf{1}_\Omega$, lattice monotonicity of the L_p norm gives

$$\| |S_{n_0} - I|(\mathbf{1}_\Omega) \|_p \geq \|S_{n_0}r_k - r_k\|_p > \frac{3}{4} \|\mathbf{1}_\Omega\|_p,$$

contradicting the choice of n_0 . Thus $L_p(\mu)$ cannot have SPAP.

Verification Notes

The proof uses only standard Banach-lattice facts:

- finite-rank operators are compact;
- compact operators send weakly null sequences to norm-null sequences;
- for a regular operator T , $|T|(|x|) \geq |Tx|$;
- the constructed dyadic signs are weakly null in finite L_p spaces.

The file `code/check_dyadic_obstruction.py` is a finite-grid sanity check for the conditional-expectation model behind the proof. It is not used as evidence for the theorem.